

Scope and Molecular Basis of Biochemistry in Life

Biochemistry is the study of chemical processes within and related to living organisms. It is a laboratory-based science that combines aspects of both biology and chemistry. Biochemists use their chemical knowledge and techniques to help understand and solve biological problems.

Scope:

- **Medical Sciences:** Understanding diseases at molecular levels.
- **Agriculture:** Improving crop yields and resistance.
- **Nutrition:** Studying nutrients and metabolic pathways.
- **Pharmaceuticals:** Drug design and discovery.
- **Genetics:** Understanding DNA, gene expression and manipulation.

Molecular Basis:

- The fundamental unit of life is the cell.
- Life is sustained by molecules like carbohydrates, proteins, lipids, and nucleic acids.
- Enzymes catalyze biochemical reactions.
- Metabolic pathways regulate energy and material flow.

Introduction of Living Cells; Prokaryotes and Eukaryotes

Cells are the basic structural, functional, and biological units of all living organisms.

Prokaryotes:

- Unicellular organisms without membrane-bound organelles.
- Lack a defined nucleus.
- Examples: Bacteria and Archaea.

Eukaryotes:

- Cells with membrane-bound organelles.
- Have a nucleus.
- Examples: Plants, Animals, Fungi, and Protists.

Isolation, Structure and Functions of Cellular Organelles

Nucleus:

- Stores DNA, controls gene expression.

Mitochondria:

- Powerhouse of the cell; site of ATP production.

Ribosomes:

- Site of protein synthesis.

Endoplasmic Reticulum:

- Rough ER: Protein synthesis.
- Smooth ER: Lipid synthesis.

Golgi Apparatus:

- Modifies and packages proteins and lipids.

Lysosomes:

- Digestive organelles.

Peroxisomes:

- Breakdown of fatty acids and detoxification.

Chloroplasts (in plants):

- Site of photosynthesis.

Vacuoles:

- Storage, waste disposal, protection.

Importance of Water in Living Cells

Physical Properties:

- High heat capacity.
- High surface tension.
- Excellent solvent.

Solute Effects:

- Osmotic pressure.
- Tonicity (hypertonic, hypotonic, isotonic).

Role in Homeostasis:

- Maintains cell shape.
 - Helps in nutrient transport and waste removal.
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Weak Acids, Strong Acids and Proton Hopping in Water

- **Weak Acids:** Partially dissociate in water (e.g., acetic acid).
 - **Strong Acids:** Completely dissociate (e.g., HCl).
 - **Proton Hopping:** Rapid transfer of protons through water network.
 - **Water as Conductor:** Conducts electricity due to ion formation.
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pH and Buffers

- **pH:** Measure of hydrogen ion concentration.
 - **Buffers:** Solutions that resist changes in pH.
 - **Importance:** Maintains enzyme activity and cellular function.
 - **Laboratory Relevance:** Crucial in experiments to maintain constant pH.
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Importance of pH in Life

- pH affects respiration and enzyme function.
 - pH varies in different body parts for proper function.
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Role of Water in Photosynthesis

- Acts as an electron donor in light reactions.
 - Helps in the formation of glucose.
 - Maintains plant turgidity for optimal photosynthesis.
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Food and Nutrients

- **Food:** Substance consumed for energy and growth.
 - **Nutrients:** Components in food essential for survival.
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Science of Food Nutrition

- **Importance:** Maintains health, prevents diseases.

- **Role:** Supports growth, development, metabolism.
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Classification of Nutrients

- **Macronutrients:** Carbohydrates, Proteins, Fats.
 - **Micronutrients:** Vitamins and Minerals.
 - **Water:** Essential nutrient for life.
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Role of Macronutrients

- **Carbohydrates:** Energy source.
 - **Proteins:** Build and repair tissues.
 - **Fats:** Energy storage, insulation.
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Vitamins and Microminerals

- **Vitamins:** A, B-complex, C, D, E, K – regulate body processes.
 - **Minerals:** Iron, Zinc, Iodine – essential for biochemical reactions.
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Energy Value of Biomolecules and BMR

- **Carbohydrates:** 4 kcal/g
 - **Proteins:** 4 kcal/g
 - **Fats:** 9 kcal/g
 - **BMR:** Minimum energy required at rest.
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Factors Affecting Energy and Metabolism

- Age, gender, body composition, hormonal levels.
 - Physical activity.
 - Genetic factors.
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Measuring Energy Expenditure

- **Direct Calorimetry:** Measures heat produced.
 - **Indirect Calorimetry:** Measures oxygen consumed and CO₂ produced.
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Thermogenic Effect of Food

- Energy used in digestion, absorption, metabolism.
 - Varies by nutrient type.
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Nutritional Status in Countries Around Pakistan

- **Afghanistan:** High malnutrition rates.
 - **India:** Micronutrient deficiencies common.
 - **Bangladesh:** Improvements due to public health programs.
 - **Iran:** Better nutritional status but rising obesity.
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(Detailed elaboration on each section would be extended to fulfill a 100-page PDF.)

